p7 review changelog (07-performance, lessons 01-24)

p7 part A: lessons 01-06 (07-performance)

Sources: react.dev reference pages (useMemo, memo, useState, use, createContext, useEffectEvent, Profiler, React Performance tracks, React Compiler intro, React 19.2 blog), web-vitals README, MDN PerformanceScriptTiming, TanStack Query important-defaults, installed zustand 5. Executed: /Users/todorbra/Projects/ReactLearning/tools/sandbox/scratch/p7a/{a,b,c}.test.tsx (Vitest + RTL, React 19.3).

All lessons: removed the warning-sign markers (verified items replaced by short confirmations; remaining hedges are plain text).

01 measure-before-optimizing

- INP thresholds: confirmed (good <=200 ms, poor >500 ms); stated explicitly. web-vitals attribution import `web-vitals/attribution`: confirmed (README). Compiler "volatile" note: confirmed stable. C2 Vitest test (memo Row, 0 re-renders over 10 parent renders): executed, 0 as claimed; `act` from `react`: confirmed. Event Timing browser support: **unresolved** (left hedged).

02 profiler-workflows

- Performance Tracks: corrected, introduced in React 19.2 (Scheduler and Components tracks; Components only for `<Profiler>` subtrees unless DevTools installed). `react-dom/profiling` alias recipe: confirmed (react.dev recommends aliasing `react-dom/client`); scheduler/tracing note reworded as obsolete. Profiler `phase`: corrected. Executed: setState in passive effect yields "update"; in layout effect yields "nested-update". Lesson previously implied effects generally gave nested-update. LoAF `scripts` field names: confirmed on MDN. LoAF Chromium-only support: not re-checked against compat data, left hedged. DevTools UI labels/compiler badge: **unresolved**.

03 why-components-rerender

- Same-state bail-out extra call: confirmed (react.dev useState + executed: repeat same-value set re-ran component once more, then stopped). B1 answer consistent. StrictMode double render: executed, 2 renders on mount. Reconciler lanes description: **unresolved** (implementation detail, hedged in text).

04 memoization-trade-offs

- useMemo cache-discard wording and 1ms guidance: confirmed. memo `arePropsEqual` deep-equality warning: confirmed. **Corrected**: claim that hand-written memoization "can often be removed gradually" after adopting the Compiler; react.dev advises leaving existing memoization in place (removal can change compiler output) or testing carefully. Compiler is stable.

05 referential-equality

- Inline ref callback behavior (React 19): confirmed by execution (no cleanup returned: attach,null,attach,null,attach; with cleanup: attach,cleanup,attach). **Corrected**: `useEffectEvent` is stable since React 19.2 (lesson said experimental). TanStack Query v5 structuralSharing default on: confirmed. Zustand `zustand/react/shallow` path: confirmed in installed package. getSnapshot caching warning: not executed, **unresolved** (unchanged).

06 context-propagation-cost

- `<Context value>` and `Context.Provider` legacy: confirmed (createContext reference). `use(Context)` conditional: confirmed. C2 test (badge 1 render, dispatch-only button 0): executed and typechecked, confirmed. Context-selector API not in stable React: still true as far as known; hedged. Dependency-list walk internals: **unresolved**.

p7 part B — lessons 07–12 (07-performance)

Method: Vitest 3 + RTL (jsdom, React 19.3.1) scratch tests in `tools/sandbox/scratch/p7b/` (a.test.tsx, b.test.tsx, c.test.tsx, d.test.tsx); MDN browser-compat-data (npm `@mdn/browser-compat-data`, fetched 2026-10-22); `web-vitals` 6.2.3 installed and typechecked; `@tanstack/react-virtual` 3.14.21 and `react-window` 2.5.0 installed; react.dev `useTransition`, React 19.2 blog; React compiler source; web.dev INP; Vite docs; all URLs in lessons 07–12 return HTTP 200.

07 State colocation

- Move-state-down / `children`-prop render counts (Heavy not re-rendered; Tbl mounts on open, unmounts on close, remounts on reopen): executed, **confirmed**.
- `<Activity>` availability: **corrected** (hedge replaced; stable in React 19.2 per blog/react.dev).
- Removed stray ⚠ / "unverified" markers (links all 200).

08 Rendering large collections

- Selectable table render count (table + 1 row), index-key input state sticking to position: executed, **confirmed**.
- `content-visibility` / `contain-intrinsic-size` support: **corrected/updated** from MDN BCD (Chrome 85/83, Firefox 125/107, Safari 18/17).
- Remaining ⚠ implementation-detail hedges softened; keys algorithm detail left hedged (**unresolved**, not testable).

09 Windowing

- TanStack Virtual v3 example: typechecked under strict TS against 3.14.21, **confirmed**; `rangeExtractor` option confirmed in typings.
- React Compiler vs TanStack Virtual: **confirmed** from compiler `DefaultModuleTypeProvider.ts` (useVirtualizer/useWindowVirtualizer marked known-incompatible); wording made specific.
- react-window: **corrected** — v2 (2.5.0) API is `List` / `Grid` with `rowComponent` / `rowCount` / `rowHeight` / `rowProps`, not FixedSizeList.
- Same-value setState bail-out: executed; first same-value set renders nothing, repeat set after a change calls the component once more: **confirmed** hedge, evidence added.
- ResizeObserver ordering (after layout, before paint): confirmed per HTML spec. Browser element-height limit and Safari momentum-scroll notes remain hedged (**unresolved**, empirical).

10 Expensive computations

- Vite `new Worker(new URL(...), {type:'module'})` + `?worker`: **confirmed** (Vite docs).
- React Compiler stable status and docs URL: **confirmed** (react.dev/learn/react-compiler and /introduction both 200).
- `toSorted` support (Chrome 110, FF 115, Safari 16), LoAF (Chromium-only, Chrome 123+): **confirmed** via BCD, stated.
- Largest-triangle-three-buckets and Comlink: markers removed (names are real); not deep-verified.

11 Scheduling

- **Corrected** support table: `scheduler.postTask` / `yield` are no longer Chromium-only (Firefox 142+; Safari none); `requestIdleCallback` Safari only flag/preview; `isInputPending` Chromium-only experimental; OffscreenCanvas support stated.
- web-vitals attribution names (`interactionTarget` , `inputDelay` , `processingDuration` , `presentationDelay` , `INPMetricWithAttribution` from `web-vitals/attribution`): typechecked on 6.2.3, **confirmed**.
- Event Timing `durationThreshold` (min 16, 8 ms rounding, default 104): confirmed per spec.
- INP thresholds 200/500 ms: **confirmed** (web.dev).
- **Corrected code bug**: C3 `useOptimistic` like-button. Executed: `setServer(await save())` after `await` is not a transition update, producing an intermediate frame with count 12 (optimistic delta on top of new server value) before 11. Fixed by wrapping post-await `setServer` / `setError` in `startTransition` (verified frames: 11,11,11 / failure 11,11,10-reverted).

12 useTransition

- Post-`await` updates not transition updates: **confirmed** (react.dev still lists as known limitation).
- **Corrected prediction** (§5.3): said `isPending` "may flip to false before they render"; executed result: batched render with `isPending=true` , then `isPending=false` .
- **Corrected**: errors thrown synchronously inside `startTransition` callback were described as normal handler exceptions; in React 19 both sync throws and rejected async actions go to the

nearest error boundary (executed + react.dev).
- Suspense in transition keeps old UI with `isPending=true` for already-revealed boundary: executed, **confirmed**; B3 memo child renders once: **confirmed**.
- React 19.2 Performance Tracks: confirmed via blog, wording updated.
- Router transitions, starvation internals, newly-mounted-boundary fallbacks: hedged (**unresolved**, implementation-dependent).

p7 part C: lessons 13-18 (07-performance)

Date 2026-10-22. Scratch: tools/sandbox/scratch/p7c/ (dv.test.tsx, lazy.test.tsx, deb.test.ts, slow.ts, vitest.config.ts).

13 useDeferredValue

- Urgent-then-background render sequence, interruptibility, no fixed delay: confirmed (react.dev/reference/react/useDeferredValue; executed RTL: `app q=ab d=a` then `d=ab`, memo child renders once with "ab"). CONFIRMED.
- `initialValue` (React 19): confirmed by docs and test (`d=""` then `d="real"`). Added: omitting it means no deferral on initial render. CONFIRMED + clarified.
- Added doc facts: an update inside a Transition returns the new value immediately (executed: `q=b d=b` in one render); Effects don't fire for background renders until commit. ADDED.
- Network note ("fewer requests than keystrokes") CORRECTED/hedged: react.dev says there is still a request per keystroke in its example; deferral does not itself prevent requests.
- Removed all warning markers.

14 Debounce and throttle

- Implementation B1/B2 hand traces: executed §3.2 code with Vitest fake timers: B1 -> [500, 950], B2 -> [0, 300, 600, 900]. CONFIRMED; note added.
- Passive listener defaults on window/document/body (Chrome, Firefox, Safari): stated more precisely (MDN-level knowledge, not re-fetched). HEDGED.
- `useEffectEvent`: no longer volatile (stable in React 19.2, react.dev); added the caveat that it may only be called from Effects. CORRECTED.
- keepalive 64 KiB quota (Fetch spec) and sendBeacon POST-only: spec text grep did not match automatically, stated from memory. UNRESOLVED (low risk).
- Scroll-driven animations support: hedged to "uneven, check MDN". UNRESOLVED exact matrix.
- Vitest timers link fixed to /guide/mocking/timers (200 OK). Removed "unverified" footer.

15 Code splitting

- `React.lazy` contract (load called on first render attempt, thenable, `.default`): react.dev + executed test (factory called once; transition keeps old UI, no fallback). CONFIRMED.
- Vite: `build.manifest` path `.vite/manifest.json` (Vite 8.3 docs) CONFIRMED. Vite 8 is Rolldown-based: `build.rollupOptions` deprecated alias of `rolldownOptions`, object

`manualChunks` removed, function form deprecated, `codeSplitting` replaces. CORRECTED (version note added in 2 places). `vite:preloadError` event added.
- Network Information `saveData`: flagged as non-Baseline/Chromium-only. CONFIRMED (MDN).
- Fixed a paren/link glitch my own edit introduced. Removed markers.

16 Bundle analysis

- rollup-plugin-visualizer 7.1.1 README: options filename/template/gzipSize/brotliSize/open/title/sourcemap/emitFile exist; templates list extended (treemap-3d, list, markdown, flamegraph); supports Rolldown, so OK with Vite 8. CONFIRMED + extended.
- size-limit: sizes are brotli by default and a plugin (`@size-limit/file`) is required; lesson labelled entries "(gz)". CORRECTED (added `"gzip": true`, install note).
- esbuild metafile `inputs[].bytes` vs `outputs[].inputs[].bytesInOutput`: executed with esbuild 0.28 (168 vs 33). CONFIRMED.
- Rollup tree-shaking anchor exists. Cache partitioning, Next `optimizePackageImports` mention (docs URL 200). Removed markers/unverified footer.

17 Network waterfalls

- `use` + promise created in render: react.dev pitfall text confirmed; wording tightened. CONFIRMED.
- HTTP/2 Server Push removed in Chrome 106: stated. 103 Early Hints wording hedged to MDN compat (not fetched in detail). HEDGED.
- Walkthrough arithmetic: HTML TTFB of 400 ms was inconsistent with "assume 50 ms server"; clarified as CDN-edge HTML. CORRECTED (downstream timings unchanged).
- Hop counts 6/3/5 re-derived by hand. CONFIRMED.

18 React vs browser bottlenecks

- Event Timing: `durationThreshold` default 104 ms, rounded to 8, min 16 (MDN). CONFIRMED.
- `interactionId` support: lesson said Chromium only; MDN compat now Chrome 96, Firefox 144, Safari 26.2 (Baseline 2025). CORRECTED.
- TypeScript 5.9 lib.dom lacks `interactionId` and `durationThreshold`; the snippet did not typecheck. CORRECTED (local `EventTimingEntry` type, cast on observe options; tsc passes).
- Long Animation Frames: Chrome/Edge 123+ only (MDN compat); `PerformanceLongTaskTiming` also Chromium-only. CORRECTED/clarified.
- grid-template-rows animation (Chrome 107, Firefox 66, Safari 16) and `transition-behavior` (Chrome 117, FF 129, Safari 17.4): from MDN BCD. CONFIRMED.
- DevTools UI claims, Lighthouse DOM-size thresholds, Selector Stats: hedged as version-dependent. UNRESOLVED (not fetched).

Out-of-scope notes for the p7 owner (lessons 11, 12, 20, 21)

- BCD data fetched 2026-10-22: `scheduler.yield` Chrome 129, Firefox 142, Safari no; `scheduler.postTask` Chrome 94, Firefox 142, Safari no; Long Animation Frames Chrome 123+

only, `presentationTime` / `paintTime` Chrome 145.
- web-vitals current version 5.x README uses onLCP/onINP/onCLS (onFID removed); attribution build via `web-vitals/attribution`. npm shows web-vitals 6.2.3 as latest; v5 README not re-checked against v6.
- Latest: react-window 2.5.0, @tanstack/react-virtual 3.14.21, vite 8.3.5.

p7 part D (lessons 19-24), reviewed 2026-10-22

Scratch: tools/sandbox/scratch/p7d/ (flush, fp, leak, b tests; Vitest 3, React 19.3.1, RTL, jsdom). Sources: web-vitals README (v6.2.3, GitHub main), web.dev (vitals, inp, fcp), MDN browser-compat-data (raw JSON), react.dev (ref callback reference, Profiler, React 19 / 19.2 / Compiler 1.0 blogs, compiler docs).

19 Memory leaks

- Ref callback cleanup semantics in React 19: CORRECTED/confirmed. react.dev: returning a cleanup means no `null` call; `null` call kept for back-compat and to be removed later. Sandbox: callback without cleanup gets `el, null`; with cleanup gets `el, cleanup`; StrictMode dev gives `el, null, el`. Lesson text updated, StrictMode note added.
- `useEffectEvent`: confirmed stable in React 19.2 (blog), exists in sandbox React 19.3.1. Marker removed.
- Strict Mode asymmetric effect shows 2 listeners, symmetric shows 1 then 0 after unmount: executed, confirmed.
- `performance.measureUserAgentSpecificMemory`: Chromium-only (MDN compat). `signal` option on addEventListener, `gcTime` naming, React 18 warning removal: confirmed/hedged wording.
- Unresolved: DevTools Memory panel labels, soak-test tooling, `performance.memory` (non-standard, left hedged).

20 Long tasks and main thread

- `scheduler.yield()`: CORRECTED. Lesson implied Chromium-only; MDN compat: Chrome 129+, Firefox 142+, Safari none. `postTask`: Chrome 94+, Firefox 142+, no Safari.
- LoAF: Chrome 123+, Chromium-only, flagged experimental: confirmed. Long Tasks (Chrome 58+ only), `isInputPending` (Chrome 87+ only, experimental), `requestIdleCallback` (Chrome 47, Firefox 55, Safari flag/preview only): confirmed.
- 50 ms threshold: confirmed. TBT definition kept; Lighthouse weight text hedged to current docs.
- Handler `setState` then `await`: executed, the "working" render runs after the sync part and before microtasks queued in the handler. Confirmed; marker replaced with sandbox note.
- Worker `new Worker(new URL(...), {type:"module"})`: hedged (Vite and webpack 5 support it).
- Unresolved: React scheduler slice length (implementation detail, left as such), compositor-vs-main-thread behavior.

21 Web vitals and budgets

- Thresholds (LCP 2.5/4 s, INP 200/500 ms, CLS 0.1/0.25, p75): confirmed via web-vitals README `*Thresholds` constants and web.dev. FCP 1.8 s confirmed. TTFB 0.8 s: unresolved (web.dev page not retrievable as text).
- CORRECTED code: `fetchpriority="high"` in JSX. React 19.3 logs "Invalid DOM property `fetchpriority`. Did you mean `fetchPriority`?" (executed with SSR and client render); changed to `fetchPriority`.
- INP selection rule: CORRECTED/hedged wording to web.dev: ignore one highest interaction per 50, then p75 across page views.
- web-vitals API: names checked against README v6.2.3: `onLCP/onINP/onCLS/onTTFB/onFCP`, attribution `target`, `resourceLoadDelay`, `resourceLoadDuration`, `elementRenderDelay`, `timeToFirstByte`, `interactionTarget`, `interactionType`, `inputDelay`, `processingDuration`, `presentationDelay`, `largestShiftTarget`, `largestShiftValue`: all confirmed. No `onFID`. Version note updated (lesson said v4/v5). INP attribution also now has `longAnimationFrameEntries`, `longestScript`.
- Soft navigations: `reportSoftNavs` opt-in, README says Chromium 151+; `navigationType` `soft-navigation`. Updated. `onCLS` Chromium-only added.
- Keepalive 64 KiB (Fetch spec) stated; React Router v7 import note; size-limit config shape hedged (not run).
- Unresolved: bfcache eligibility rules, DevTools throttling presets, `deviceMemory` / `connection` support (left as Chromium-only per memory).

22 Production performance investigations

- `responseStatus` on resource timing: Chrome 109+, Firefox 129+, not Safari (MDN). `serverTiming`: Chrome 65, Firefox 61, Safari 16.4. Marker replaced with these facts.
- Process/Amdahl content is conceptual; no executable claims.

23 Counterproductive optimizations

- React Compiler: CORRECTED/updated. Stable v1.0 released 2025-10-07 (blog; npm babel-plugin-react-compiler 1.0.0); React 17+ with `react-compiler-runtime`; lint rules in eslint-plugin-react-hooks recommended preset; "Memo ✨" DevTools badge; `"use no memo"` directive: all confirmed on react.dev. The "verify" banner replaced by a status note.
- Profiler disabled in production builds unless profiling build: confirmed on react.dev Profiler reference.
- Unresolved: the experiment magnitudes in section 4 (illustrative, not measured); prefetch hit-rate threshold left as heuristic.

24 Phase assessment

- Executed all render-count/ordering answers: B1 (400 / 100), B2 (3 with and without memo), B3 (`S:ab/a S:ab/ab R:ab`, SearchPage 2, Results 1 with "ab"), B4 (4/1/1), B5 (`A E G C D F B`, node), B7 (3 intervals, none cleared, 3 timers remain after unmount, fake timers): all CONFIRMED, no change.

- Not executed: B6 (hop counting, reasoned), B8 (layout thrash counts, needs a browser), B9, B10, Parts C and D numerics.
- Updated: `scheduler.yield` support, Web Vitals thresholds (verified), `useEffectEvent` stable in 19.2, compiler now stable v1.0.
- Unresolved: Part D implementation answers not run.